

1. Cyber Crimes

Cyber crime refers to illegal activities that involve a computer, networked device, or a network. These crimes use cyberspace as either a tool or target, involving data theft, identity theft, hacking, fraud, cyberstalking, and more.

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Types of Cyber Crimes: Hacking, Phishing, Identity theft, Cyberstalking, Online fraud, Malware attacks, Ransomware, Denial of Service (DoS) attacks.

2. Cyber Crimes in the Context of Internet

The internet provides anonymity and global connectivity, increasing cyber crime risks. Cyber criminals exploit vulnerabilities in systems, networks, or human behavior.

3. Legal Aspects of Open Communications

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Communications over the internet are subject to laws governing privacy, interception, data protection, and misuse.

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Open communications pose challenges regarding **jurisdiction**, **evidence collection**, and **data privacy**.

4. Indian Penal Law & Cyber Crimes

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IT Act, 2000: Primary legislation addressing cyber crimes in India.

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Sections cover hacking (Section 66), data theft (Section 43), identity theft, cyber terrorism (Section 66F), and others.

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Offenses like **fraud**, **mischief**, and **defamation** can also apply when committed online.

5. International Law on Cyber Crimes

- Cyber crimes are global; hence, international cooperation is vital.
- **Budapest Convention on Cybercrime** is the first international treaty addressing crimes committed via the internet and other computer networks.
- Countries have their own laws but cooperate on extradition, investigation, and prosecution.

6. Obscenity and Pornography on the Internet

- Laws regulate the transmission and hosting of obscene or pornographic material.
- In India, sections under IT Act and Indian Penal Code (IPC) address **publishing or transmitting obscene material**.
- Controversy exists regarding freedom of expression vs. societal norms.

7. Introduction to Ethical Hacking

- **Ethical hacking or penetration testing** is authorized probing of systems to identify vulnerabilities before malicious hackers exploit them.
- Ethical hackers use the same tools and techniques as malicious hackers but report findings responsibly.

8. Ethical Hacking Terminology

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Vulnerability: Weakness in a system that can be exploited.

- **Exploit:** Code or technique that takes advantage of a vulnerability.
- **Payload:** Malicious code delivered by an exploit.
- **Backdoor:** Secret method to bypass security controls.
- **Reconnaissance:** Initial phase of gathering information about the target.
- **Footprinting:** Collecting network and system details.
- **Scanning:** Actively probing a system for vulnerabilities.
- **Privilege escalation:** Gaining higher access rights than authorized.

9. Types of Hacking Technologies

- **Black Hat:** Malicious hackers breaking security for personal gain.
- **White Hat:** Ethical hackers authorized to test security.
- **Gray Hat:** Hackers who may violate laws but without malicious intent.
- **Script Kiddies:** Novices using pre-made tools.
- **Hacktivists:** Hackers with political or social agendas.

- **State-sponsored hackers:** Employed by governments for cyber espionage.

10. Phases Involved in Ethical Hacking

- **Reconnaissance:** Passive and active information gathering.
- **Scanning:** Network scanning using tools like Nmap.
- **Gaining Access:** Exploiting vulnerabilities.
- **Maintaining Access:** Installing backdoors or Trojans.
- **Covering Tracks:** Clearing logs and footprints.